Colon & Colorectal Cancer — FDA Approved Drugs Intelligence

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Situation Summary

Biomarker: 16 require biomarker testing
Immunotherapy: 3 immunotherapy agents
Targeted: 16 biomarker-directed agents
Breakthrough: 16 drugs scored ≥ 4

FDA Approved Drug Timeline

FDA-approved colorectal cancer drugs ranked by clinical significance. Drugs requiring biomarker testing are marked with the required biomarker. Click any drug name to view the full FDA approval record.

HIGH IMMUNOTHERAPY REQUIRES: MSI-H/dMMR FDA APPROVED 20250919 5

Pembrolizumab (Keytruda)

Pembrolizumab, marketed as Keytruda, is approved for the treatment of colorectal cancer in patients with microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) tumors. This drug functions as an immunotherapy by blocking the programmed death receptor-1 (PD-1), which helps to enhance the immune system's ability to detect and attack cancer cells. Before considering pembrolizumab, patients should be aware of the specific biomarker requirement of MSI-H/dMMR, as this determines eligibility for treatment. Pembrolizumab is part of a class of immunotherapies that target the PD-1 pathway, which may differ structurally from other treatment options available for colorectal cancer.

HIGH RET-TARGETED REQUIRES: RET fusion FDA APPROVED 20240410 5

Selpercatinib (Retevmo)

Selpercatinib, marketed as Retevmo, is approved for the treatment of adult patients with locally advanced or metastatic colorectal cancer that harbors a RET fusion, representing a targeted therapy option. As a RET inhibitor, the drug specifically targets the RET gene fusion, which is crucial for determining eligibility for treatment. Patients should be aware that testing for the RET fusion is necessary before considering this drug, as it is only effective in those with the specific biomarker.

HIGH NTRK-TARGETED REQUIRES: NTRK fusion FDA APPROVED 20231020 5

Entrectinib (Rozlytrek)

Entrectinib, marketed as Rozlytrek, is approved for the treatment of adult and pediatric patients with solid tumors that have a neurotrophic tyrosine receptor kinase (NTRK) gene fusion, including colorectal cancer, as detected by an FDA-approved test. This drug functions as a TRK inhibitor, targeting tumors with the NTRK fusion, which is significant as it can drive cancer growth. Before considering this drug, patients should be aware of the necessity for testing to confirm the presence of the NTRK fusion and the absence of known acquired resistance mutations. Rozlytrek is structurally distinct from other kinase inhibitors, providing a targeted approach to treatment based on specific genetic alterations.

HIGH KRAS-TARGETED REQUIRES: KRAS G12C FDA APPROVED 20221212 5

Adagrasib (Krazati)

Adagrasib, marketed as Krazati, is approved for the treatment of adult patients with KRAS G12C-mutated locally advanced or metastatic colorectal cancer (CRC) in combination with cetuximab. This drug functions as a KRAS G12C inhibitor, targeting a specific mutation that plays a critical role in cancer cell growth and survival. Before considering this drug, patients should confirm the presence of the KRAS G12C mutation through an FDA-approved test, as this biomarker is essential for treatment eligibility. Krazati is part of a class of targeted therapies that specifically inhibit the KRAS GTPase family, which may differ structurally from other treatment options available for CRC.

HIGH KRAS-TARGETED REQUIRES: KRAS G12C FDA APPROVED 20210528 5

Sotorasib (Lumakras)

Sotorasib, marketed as Lumakras, is approved for the treatment of adult patients with KRAS G12C-mutated colorectal cancer who have received at least one prior systemic therapy. As a KRAS G12C inhibitor, it specifically targets the KRAS G12C mutation, which is a critical biomarker for determining eligibility for this therapy. Patients should be aware that the presence of the KRAS G12C mutation must be confirmed through an FDA-approved test before considering this drug. Lumakras is distinct in its mechanism compared to other therapies targeting different mutations or pathways in colorectal cancer.

HIGH HER2-TARGETED REQUIRES: HER2+ FDA APPROVED 20200417 5

Tucatinib (Tukysa)

Tukysa (tucatinib) is approved for the treatment of adult patients with RAS wild-type HER2-positive unresectable or metastatic colorectal cancer. As a HER2 inhibitor, Tukysa targets the HER2 protein, which is overexpressed in some colorectal cancers, making the biomarker HER2 status critical for determining eligibility for this therapy. Before considering Tukysa, patients should discuss their HER2 status and any previous treatments with their healthcare provider to ensure it aligns with the drug's indications. Tukysa is part of a class of HER2 inhibitors, which may offer different structural properties compared to other available options.

HIGH HER2-TARGETED REQUIRES: HER2+ FDA APPROVED 20191220 5

Trastuzumab deruxtecan (Enhertu)

Trastuzumab deruxtecan, marketed as Enhertu, is approved for the treatment of HER2-positive metastatic colorectal cancer in patients who have received prior therapies. This drug functions as an antibody-drug conjugate (ADC) that targets the HER2 protein, which is significant as the presence of the HER2 biomarker can influence treatment efficacy. Before considering this drug, patients should be aware of the requirement for HER2 positivity, as determined by an FDA-approved test, to ensure appropriate use. Enhertu is structurally distinct from traditional chemotherapy options, providing a targeted approach to treatment.

HIGH HER2-TARGETED REQUIRES: HER2+ FDA APPROVED 20190228 5

Trastuzumab (Herceptin)

Trastuzumab, marketed as Herceptin, is approved for the treatment of HER2-overexpressing colorectal cancer in adults. It functions as a HER2/neu receptor antagonist, targeting the HER2 biomarker, which is crucial for identifying patients who may benefit from this therapy. Before considering Herceptin, patients should be tested for HER2 overexpression to ensure they are eligible for treatment. This drug is part of a class of anti-HER2 therapies, which specifically target the HER2 protein involved in cancer cell growth.

HIGH NTRK-TARGETED REQUIRES: NTRK fusion FDA APPROVED 20181126 5

Larotrectinib (Vitrakvi)

Vitrakvi (larotrectinib) is approved for the treatment of adult and pediatric patients with colorectal cancer that has an NTRK gene fusion and is either metastatic or where surgical resection may lead to severe morbidity. As a TRK inhibitor, Vitrakvi targets tumors with NTRK fusions, which are critical for determining eligibility for this therapy. Before considering Vitrakvi, patients should be tested for the presence of the NTRK fusion, as this biomarker is essential for treatment selection. Vitrakvi is structurally distinct from other therapies in its class, focusing specifically on the inhibition of TRK proteins associated with NTRK fusions.

HIGH BRAF-TARGETED REQUIRES: BRAF V600E FDA APPROVED 20180627 5

Encorafenib (Braftovi)

Encorafenib, marketed as Braftovi, is approved for the treatment of adult patients with metastatic colorectal cancer (mCRC) in combination with cetuximab and fluorouracil-based chemotherapy, specifically for those with a BRAF V600E mutation. This drug functions as a BRAF inhibitor, targeting the BRAF V600E mutation, which is significant as it plays a role in the growth and spread of cancer cells. Before considering this drug, patients should be aware that testing for the BRAF V600E mutation is required to determine eligibility for treatment. Braftovi is part of a class of targeted therapies that focus on specific genetic mutations, differentiating it from traditional chemotherapy options.

HIGH BRAF-TARGETED REQUIRES: BRAF V600E FDA APPROVED 20180627 5

Binimetinib (Mektovi)

Mektovi (binimetinib) is approved for the treatment of patients with unresectable or metastatic colorectal cancer that has a BRAF V600E mutation, in combination with encorafenib. As a MEK inhibitor, it works by targeting the MAPK signaling pathway, which is often activated in cancers with BRAF mutations, making the biomarker critical for identifying eligible patients. Before considering Mektovi, patients should discuss with their healthcare provider whether they have the BRAF V600E mutation, as this determines the drug's appropriateness. Mektovi is structurally distinct from other kinase inhibitors, providing a targeted approach in the treatment of specific mutations.

HIGH IMMUNOTHERAPY REQUIRES: MSI-H/dMMR FDA APPROVED 20150304 5

Nivolumab (Opdivo)

Nivolumab, marketed as Opdivo, is approved for the treatment of colorectal cancer in patients with microsatellite instability-high (MSI-H) or mismatch repair-deficient (dMMR) tumors, representing a subsequent line of therapy. This drug functions as an immunotherapy by blocking the programmed death receptor-1 (PD-1), which helps enhance the immune system's ability to recognize and attack cancer cells, making the biomarker status critical for identifying eligible patients. Before considering this drug, patients should discuss their MSI-H/dMMR status with their healthcare provider, as this determines the appropriateness of nivolumab for their treatment. Nivolumab is part of a class of PD-1 inhibitors, which includes other options that may have different indications and mechanisms of action.

HIGH HER2-TARGETED REQUIRES: HER2+ FDA APPROVED 20120608 5

Pertuzumab (Perjeta)

Pertuzumab, marketed as Perjeta, is approved for the treatment of HER2-positive colorectal cancer in combination with trastuzumab and chemotherapy. It functions as a HER2/neu receptor antagonist, targeting the HER2 biomarker, which is crucial for identifying patients who may benefit from this therapy. Before considering this drug, patients should discuss their HER2 status with their healthcare provider, as it is a requirement for treatment eligibility. Pertuzumab is structurally similar to other anti-HER2 agents, enhancing its effectiveness when used in combination therapies.

HIGH IMMUNOTHERAPY REQUIRES: MSI-H/dMMR FDA APPROVED 20110325 5

Ipilimumab (Yervoy)

Ipilimumab, marketed as Yervoy, is an FDA-approved immunotherapy for the treatment of colorectal cancer in patients with microsatellite instability-high (MSI-H) or mismatch repair-deficient (dMMR) tumors. This drug functions as a CTLA-4-blocking antibody, enhancing the immune system's ability to target and destroy cancer cells, which is particularly relevant for patients with the specified biomarkers. Before considering Yervoy, patients should discuss potential side effects and the importance of biomarker testing with their healthcare provider.

HIGH ANTI-EGFR REQUIRES: RAS wild-type FDA APPROVED 20060927 5

Panitumumab (Vectibix)

Panitumumab, marketed as Vectibix, is an FDA-approved anti-EGFR agent indicated for the treatment of adult patients with wild-type RAS metastatic colorectal cancer (mCRC). It is used in combination with FOLFOX as a first-line treatment and as a monotherapy after disease progression following prior chemotherapy. Before considering Vectibix, patients should ensure they have been tested for RAS mutations, as the drug is only effective in those with wild-type RAS.

Structurally, Vectibix is an epidermal growth factor receptor (EGFR) antagonist, similar to other agents in its class but specifically targeting the RAS wild-type population.

HIGH ANTI-EGFR REQUIRES: RAS wild-type FDA APPROVED 20040212 5

Cetuximab (Erbitux)

Cetuximab, marketed as Erbitux, is approved for the treatment of RAS wild-type metastatic colorectal cancer, typically used in combination with chemotherapy as a first-line therapy. This drug functions as an epidermal growth factor receptor (EGFR) antagonist, targeting cancer cells that express the EGFR, which is crucial for tumor growth and survival. Before considering this drug, patients should be tested for RAS mutations, as the presence of RAS mutations indicates that cetuximab may not be effective. Structurally, cetuximab is a monoclonal antibody, distinguishing it from other treatments that may target different pathways or receptors in cancer therapy.

MEDIUM ANTI-EGFR FDA APPROVED 20231108 3

Fruquintinib (Fruzaqla)

Fruquintinib, a VEGFR inhibitor, is indicated for adult patients with metastatic colorectal cancer (mCRC) who have previously received fluoropyrimidine-, oxaliplatin-, and irinotecan-based chemotherapy, an anti-VEGF therapy, and, if RAS wild-type and medically appropriate, an anti-EGFR therapy. No biomarker requirement is specified.

MEDIUM CHEMOTHERAPY FDA APPROVED 20150922 2

Trifluridine (Lonsurf)

Lonsurf, a chemotherapy combination of trifluridine and tipiracil, is indicated for adult patients with metastatic colorectal cancer who have previously received fluoropyrimidine-, oxaliplatin-, and irinotecan-based therapies. It can be used as a single agent or in combination with bevacizumab. There are no biomarker requirements for its use.

MEDIUM ANTI-EGFR FDA APPROVED 20140421 2

Ramucirumab (Cyramza)

Ramucirumab (Cyramza), an anti-VEGFR2 agent, is indicated for colorectal cancer in combination with FOLFIRI for patients with metastatic colorectal cancer who have disease progression after prior therapy. There is no biomarker requirement for its use.

MEDIUM MULTI-KINASE FDA APPROVED 20120927 2

Regorafenib (Stivarga)

Regorafenib, a multi-kinase inhibitor, is indicated for metastatic colorectal cancer in adult patients who have previously received fluoropyrimidine-, oxaliplatin-, and irinotecan-based chemotherapy, an anti-VEGF therapy, and, if RAS wild-type, an anti-EGFR therapy. No biomarker requirement is specified.

MEDIUM ANTI-VEGF FDA APPROVED 20120803 2

Aflibercept (Zaltrap)

Aflibercept, an anti-VEGF agent, is indicated for metastatic colorectal cancer (mCRC) in combination with FOLFIRI for patients whose disease is resistant to or has progressed after an oxaliplatin-containing regimen. There is no biomarker requirement for its use.

MEDIUM CHEMOTHERAPY FDA APPROVED 20050131 2

Oxaliplatin (Eloxatin)

Oxaliplatin, a chemotherapy agent, is indicated for the treatment of colorectal cancer. It does not have a biomarker requirement and is utilized in various lines of therapy for this indication.

MEDIUM ANTI-VEGF FDA APPROVED 20040226 ⚡ 2

Bevacizumab (Avastin)

Bevacizumab (Avastin), an anti-VEGF agent, is indicated for metastatic colorectal cancer. It is used in combination with intravenous fluorouracil-based chemotherapy for first- or second-line treatment, and with fluoropyrimidine-irinotecan or fluoropyrimidine-oxaliplatin-based chemotherapy for second-line treatment. No biomarker requirement is specified.

MEDIUM CHEMOTHERAPY FDA APPROVED 19980430 ⚡ 2

Capecitabine (Xeloda)

Capecitabine (Xeloda), a nucleoside metabolic inhibitor, is indicated for the adjuvant treatment of Stage III colon cancer as a single agent or in combination chemotherapy. It is also approved for the perioperative treatment of adults with locally advanced rectal cancer as part of chemoradiotherapy. No biomarker requirement is specified for its use in colorectal cancer.

MEDIUM CHEMOTHERAPY FDA APPROVED 19960614 ⚡ 2

Irinotecan (Camptosar)

Irinotecan, a chemotherapy agent, is indicated for metastatic colorectal carcinoma as part of first-line therapy in combination with 5-fluorouracil and leucovorin. It is also approved for patients whose disease has recurred or progressed following initial fluorouracil-based therapy. No biomarker requirement is specified.

MEDIUM CHEMOTHERAPY ⚡ 2

Fluorouracil (5-FU)

Fluorouracil (5-FU), a chemotherapy agent, is indicated for the treatment of colorectal cancer. It does not have a biomarker requirement and can be utilized in various lines of therapy for this indication.

Appendix A — Patient Guide to Approved Drugs

For patients, caregivers, and family members. Always discuss treatment options with your oncologist — which drug is right for you depends on your tumor's molecular profile, stage, and prior treatment history.

Why biomarker testing matters before treatment

Many FDA-approved colorectal cancer drugs only work — or are only approved — for tumors with a specific genetic mutation. Getting complete biomarker testing at diagnosis (RAS, BRAF, MSI/MMR, HER2, NTRK) ensures you know which of these drugs you may qualify for. Testing should happen before your first treatment begins.

Pembrolizumab (Keytruda)
IMMUNOTHERAPY · Requires: MSI-H/dMMR
Pembrolizumab, marketed as Keytruda, is approved for the treatment of colorectal cancer in patients with microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR) tumors. This drug functions as an immunotherapy by blocking the programmed death receptor-1 (PD-1), which helps to enhance the immune system's ability to detect and attack cancer cells. Before considering pembrolizumab, patients should be aware of the specific biomarker requirement of MSI-H/dMMR, as this determines eligibility for treatment. Pembrolizumab is part of a class of immunotherapies that target the PD-1 pathway, which may differ structurally from other treatment options available for colorectal cancer.

Selpercatinib (Retevmo)
RET-TARGETED · Requires: RET fusion
Selpercatinib, marketed as Retevmo, is approved for the treatment of adult patients with locally advanced or metastatic colorectal cancer that harbors a RET fusion, representing a targeted therapy option. As a RET inhibitor, the drug specifically targets the RET gene fusion, which is crucial for determining eligibility for treatment. Patients should be aware that testing for the RET fusion is necessary before considering this drug, as it is only effective in those with the specific biomarker.

Entrectinib (Rozlytrek)
NTRK-TARGETED · Requires: NTRK fusion
Entrectinib, marketed as Rozlytrek, is approved for the treatment of adult and pediatric patients with solid tumors that have a neurotrophic tyrosine receptor kinase (NTRK) gene fusion, including colorectal cancer, as detected by an FDA-approved test. This drug functions as a TRK inhibitor, targeting tumors with the NTRK fusion, which is significant as it can drive cancer growth. Before considering this drug, patients should be aware of the necessity for testing to confirm the presence of the NTRK fusion and the absence of known acquired resistance mutations. Rozlytrek is structurally distinct from other kinase inhibitors, providing a targeted approach to treatment based on specific genetic alterations.

Adagrasib (Krazati)
KRAS-TARGETED · Requires: KRAS G12C
Adagrasib, marketed as Krazati, is approved for the treatment of adult patients with KRAS G12C-mutated locally advanced or metastatic colorectal cancer (CRC) in combination with cetuximab. This drug functions as a KRAS G12C inhibitor, targeting a specific mutation that plays a critical role in cancer cell growth and survival. Before considering this drug, patients should confirm the presence of the KRAS G12C mutation through an FDA-approved test, as this biomarker is essential for treatment eligibility. Krazati is part of a class of targeted therapies that specifically inhibit the KRAS GTPase family, which may differ structurally from other treatment options available for CRC.

Tucatinib (Tukysa)
HER2-TARGETED · Requires: HER2+
Tukysa (tucatinib) is approved for the treatment of adult patients with RAS wild-type HER2-positive unresectable or metastatic colorectal cancer. As a HER2 inhibitor, Tukysa targets the HER2 protein, which is overexpressed in some colorectal cancers, making the biomarker HER2 status critical for determining eligibility for this therapy. Before considering Tukysa, patients should discuss their HER2 status and any previous treatments with their healthcare provider to ensure it aligns with the drug's indications. Tukysa is part of a class of HER2 inhibitors, which may offer different structural properties compared to other available options.

Encorafenib (Braftovi)
BRAF-TARGETED · Requires: BRAF V600E
Encorafenib, marketed as Braftovi, is approved for the treatment of adult patients with metastatic colorectal cancer (mCRC) in combination with cetuximab and fluorouracil-based chemotherapy, specifically for those with a BRAF V600E mutation. This drug functions as a BRAF inhibitor, targeting the BRAF V600E mutation, which is significant as it plays a role in the growth and spread of cancer cells. Before considering this drug, patients should be aware that testing for the BRAF V600E mutation is required to determine eligibility for treatment. Braftovi is part of a class of targeted therapies that focus on specific genetic mutations, differentiating it from traditional chemotherapy options.

Panitumumab (Vectibix)

ANTI-EGFR · Requires: RAS wild-type

Panitumumab, marketed as Vectibix, is an FDA-approved anti-EGFR agent indicated for the treatment of adult patients with wild-type RAS metastatic colorectal cancer (mCRC). It is used in combination with FOLFOX as a first-line treatment and as a monotherapy after disease progression following prior chemotherapy. Before considering Vectibix, patients should ensure they have been tested for RAS mutations, as the drug is only effective in those with wild-type RAS. Structurally, Vectibix is an epidermal growth factor receptor (EGFR) antagonist, similar to other agents in its class but specifically targeting the RAS wild-type population.

Trifluridine (Lonsurf)

CHEMOTHERAPY ·

Lonsurf, a chemotherapy combination of trifluridine and tipiracil, is indicated for adult patients with metastatic colorectal cancer who have previously received fluoropyrimidine-, oxaliplatin-, and irinotecan-based therapies. It can be used as a single agent or in combination with bevacizumab. There are no biomarker requirements for its use.

Regorafenib (Stivarga)

MULTI-KINASE ·

Regorafenib, a multi-kinase inhibitor, is indicated for metastatic colorectal cancer in adult patients who have previously received fluoropyrimidine-, oxaliplatin-, and irinotecan-based chemotherapy, an anti-VEGF therapy, and, if RAS wild-type, an anti-EGFR therapy. No biomarker requirement is specified.

Aflibercept (Zaltrap)

ANTI-VEGF ·

Aflibercept, an anti-VEGF agent, is indicated for metastatic colorectal cancer (mCRC) in combination with FOLFIRI for patients whose disease is resistant to or has progressed after an oxaliplatin-containing regimen. There is no biomarker requirement for its use.

What this means for you

Based on the drugs in this report, **7 distinct biomarker profiles** determine access to targeted therapies: **MSI-H/dMMR, RET fusion, NTRK fusion, KRAS G12C, HER2+, BRAF V600E, and RAS wild-type**. If your tumor has not been tested for these markers, ask your oncologist before your next treatment decision. Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and determine which of these FDA-approved drugs you may qualify for.

Questions to ask your oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"
"Which of these approved drugs apply to my specific mutation profile?"
"What line of therapy is each drug approved for — first-line, second-line, or later?"

Source: FDA Drug@FDA · 2026-04-25 23:34:00
This is not medical advice. Always speak with your oncologist before making any treatment decisions.

Appendix B — CRC Drug Reference Guide

Biomarker-Directed Therapies

MSI-H/dMMR — Pembrolizumab (Keytruda), Nivolumab (Opdivo) + Ipilimumab (Yervoy)

BRAF V600E — Encorafenib (Braftovi) + Cetuximab or Binimetinib

HER2+ — Tucatinib (Tukysa), Trastuzumab deruxtecan (Enhertu), Pertuzumab + Trastuzumab

NTRK fusion — Larotrectinib (Vitrakvi), Entrectinib (Rozlytrek)

KRAS G12C — Sotorasib (Lumakras), Adagrasib (Krazati)

RAS wild-type — Cetuximab (Erbitux), Panitumumab (Vectibix) [anti-EGFR agents]

Chemotherapy Backbones

FOLFOX — Fluorouracil + Leucovorin + Oxaliplatin

FOLFIRI — Fluorouracil + Leucovorin + Irinotecan

FOLFOXIRI — All three combined; used in high-volume metastatic disease

Capecitabine (Xeloda) — Oral fluorouracil alternative

Trifluridine/tipiracil (Lonsurf) — Later-line oral chemotherapy

Anti-Angiogenic Agents

Bevacizumab (Avastin) — Anti-VEGF; added to FOLFOX or FOLFIRI in first-line

Ramucirumab (Cyramza) — Anti-VEGFR2; second-line with FOLFIRI

Aflibercept/Ziv-aflibercept (Zaltrap) — Anti-VEGF; second-line

Fruquintinib (Fruzaqla) — VEGFR inhibitor; later-line, no biomarker required

Regorafenib (Stivarga) — Multi-kinase inhibitor; later-line

Key principle

Most biomarker-directed drugs require confirmed mutation status before use. RAS and BRAF testing is required before starting anti-EGFR agents. MSI/MMR testing determines immunotherapy eligibility. Always test before starting first-line therapy — not after.

Key Resources

FDA Drug Approvals (accessdata.fda.gov) · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org)

Colon & Colorectal Cancer — FDA Approved Drugs Intelligence

About This Report

This report is generated automatically from publicly available medical databases including PubMed, ClinicalTrials.gov, and the FDA Drug@FDA database. Entries are classified using deterministic rule-based logic and scored by clinical significance. AI-generated summaries are grounded exclusively in the source data provided.

Limitations

This report does not represent a complete picture of all published research or approved treatments. Database coverage, feed availability, and relevance filtering affect what appears. New drug approvals and trial status changes may not be reflected immediately. Always verify current information at the original source before making clinical decisions.

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